MILITARY SPECIFICATION

FRONT SIGHT ASSEMBLY

1. SCOPE

1.1 This specification covers special requirements not normally required by the drawings and provides quality assurance provisions for the fabrication of one type of front sight designated as Front Sight Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

UU-T-106 - Tape pressure Sensitive Adhesive Masking Paper.

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-E-5272 - Environmental testing, Aeronautical Equipment, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.
MIL-P-17667 - Paper wrapping chemically neutral.

STANDARDS

FEDERAL

Federal Test Method Std No. 141 - Paint, Varnish and Related Materials; method of Inspection, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.

FSC: 1340

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MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, packing and marking for shipment of Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

9234281 - Front Sight Assembly.

PUBLICATIONS

ORDNANCE CORPS

ET 9234281 - Equipment Tabulation

OTHER

Code of Federal Regulation, 49, 71-90
National Bureau of Standards Handbook No. 66 - Safe Design and use of Industrial Beta Ray Sources.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Front Sight Assembly. - The sight shall comply with all requirements specified on Drawing 9234281 and with all requirements specified in applicable specifications.

3.3 Carrier - The inert spherical ceramic material for containing the Promethium 147 (Pm-147) radioisotope shall be identified as the carrier and shall be a dry, free flowing material with a diameter in the range of 5 microns minimum to 60 microns maximum.

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3.4 Activator - The Pm-147 radioisotope when housed in the carrier shall be identified as the activator.

3.4.1 Activator specific activity level - The activator shall have a minimum specific activity of 100 millicuries per gram of activator.

3.4.2 Activator solubility - The activator shall exhibit a solubility not in excess of 0.01 percent when tested as specified in 4.4.1.2.

3.4.3 Activator temperature stability.-The activator shall withstand conditioning in air at 1800 degrees Fahrenheit (F) plus or minus 10 degrees F for one hour with less than 1.0 percent release of Pm-147.

3.4.4 Activator acid resistance - The activator shall exhibit a solubility not in excess of 1.0 percent when a sample of the activator is soaked in 6 normal hydrochloric acid for a minimum of 25 hours at 25 degrees centigrade (C) as specified in 4.4.1.4.

3.5 Cured Pm-147 activated self-luminous paint.

3.5.1 Wipe test for radiological contamination from cured self-luminous paint. - Not more than 2,200 disintegrations per minute shall be released from cured Pm-147 activated self-luminous paint when it is wipe tested as specified in 4.4.2.1.

3.5.2 Solubility of cured self-luminous paint - The cured Pm-147 activated self-luminous paint shall exhibit a solubility not in excess of 0.01 percent of the total Pm-147.

3.5.3 Consistency and level of radioactive material in cured self-luminous paint - The cured Pm-147 activated self-luminous paint shall be a uniform mixture of phosphorescent and activator materials and binder. The maximum amount of radioactive material within the sight shall be 3 millicuries.

3.6 Sight.-The sight shall be composed of pieces of clear plastic which are joined together in a clear plastic embediment resulting in a sealed source. All sealing shall be internal and no adhesive seal will be permited on any external surface of the finished casting or embediment. The completed sight shall be clear, transparent and colorless.

3.7 Complete sight performance and safety.

3.7.1 Phosphorescent color of the complete sight - The color of the luminous marking shall have a dominant wavelength between 5,400 and 5,600 angstroms. The dominent wavelength shall be determined as specified in 4.4.3.1.

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3.7.2 Brightness of the complete sight.-The self-luminous paint shall exhibit a uniform brightness between 80 microlamberts minimum and 100 microlamberts maximum when measured after storage in complete darkness for 1 day minimum immediately after manufacture.

3.7.2.1 Brightness decay - The brightness of each self-luminous paint marking shall at 28 months be not less than 50 percent of the brightness one day after manufacture when tested as specified herein.

3.7.3 Radiation contamination of the complete sight. - When the sight is subjected to both the radiological immersion and the wipe tests specified in 4.4.3.3, the complete sight shall not release more than 2,200 disintegrations per minute from either test.

3.7.4 Radiation level of complete sight.-The level of radiation emitted from any surface of the finished sight shall not exceed .5 millirad per hour at 10 centimeters when measured through 50 milligrams per square centimeter of absorber.

3.8 Environmental conditioning tests of complete sight.-The complete sight shall be capable of withstanding the environmental conditions specified in 4.4.4 without deterioration to the extent that following the environmental conditioning testing and without wiping or cleaning, the complete sight shall comply with the following requirements.

3.8.1 Damage.-The complete sight shall show no water leakage, cracking, chipping, flaking, separation of laminations, or damage other than surface scratches.

3.8.2 Brightness decay following environmental conditioning.-The complete sight shall comply with the brightness decay portion of the brightness requirement as specified in 3.7.2.

3.8.3 Radioactive contamination following environmental conditioning.-The complete sight shall comply with the radiation contamination requirement specified in 3.7.3.

3.9 Workmanship.-All parts, subassemblies and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter that may impair proper functioning of the sight or that may cause deterioration in storage other than brightness decay deterioration as described in paragraph 3.7.2. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent used. Painted or coated surfaces shall be restored when these surfaces have been marred or scratched. All required markings shall be neat, clean, and sharply defined.

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3.10 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of the following:

a. 25 complete sights

b. 0.30 gram to 0.70 gram sample of activator per activator batch.

c. 5 Samples from each cured Pm-147 activated luminous paint batch as specified in paragraph 4.3.3.1.2.1.

The first article sample shall consist of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials, (see 4.1.1.e). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, materials used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. - Each article in the first article sample shall be examined and inspected for every dimension and requirement on the drawing. The entire sample shall be examined in accordance with 4.3.2 and then tested as specified in 4.3.3.1.

4.2.3 Rejection. - The first article sample shall be rejected if any assembly or component fails to conform to any or all of the applicable requirements. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample, or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the Contracting Officer.

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4.3 Inspection provisions.

4.3.1 Lot formation. - The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, specifications, or revision thereof. Changes to the process, specification, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain paint from not more than one mixture of activator pigment and binder, the mixture of which shall be defined as a paint batch. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. - Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects (and majors, when so specified) shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Sight, Front (see 3.8.1 and Dwg. 9234282).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Vertical distance between range graduations	Gage	01001
102.	Distance of centerline from edge	Gage	01002
Minor:	AQL 1.50 percent		
201.	Identification marking not legible or complete	Visual	01003
202.	Height to 200 meter mark	Gage	01004
203.	Location of stadia lines	Gage	01005
204.	Evidence of poor workmanship	Visual	01006

4.3.2.2 Sight, Front Assembly (see Dwg. 9234281 and 3.8.1).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Sight contains physical visual defect or damage that could affect contamination by the self-luminous paint (see 3.8.1)	Visual	02001
Major:	AQL 1.00 percent		
101.	Material not clear	Visual	02002
102.	Sight material produces visual distortion	Visual	02003
103.	Sight marking not legible or complete	Visual	02004
104.	Luminous portion not complete	Visual	02005
105.	Improper Adhesion (including bubbles)	Visual	02006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship	Visual	02007

4.3.3 Testing

4.3.3.1 First Article Sample (see 4.2.1). - The samples submitted as the first article quantity shall be tested as specified herein after being examined 100 percent for the applicable classifications of defects in Paragraph 4.3.2.

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4.3.3.1.1 Activator

4.3.3.1.1.1 Activator specific activity level (see 3.4.1 and 4.4.1.1) Major defect - Code No. 03001 - a 0.3 gram minimum sample of activator shall be selected from each activator batch prepared to produce the first article sample. The sample shall be tested as specified in 4.4.1.1 using test equipment specified in 4.3.4. The first article sample shall be rejected if it contains an activator batch that is classed defective.

4.3.3.1.1.2 Activator solubility (see 3.4.2 and 4.4.1.2) Critical defect - Code No. 03002 - Five samples of activator weighing between 0.02 grams to 0.10 grams shall be selected from each activator batch prepared to produce the first article sample. The samples shall be tested as specified in 4.4.1.2 using test equipment specified in 4.3.4. The first article sample shall be rejected if it contains an activator batch that is classed defective.

4.3.3.1.1.3 Activator temperature stability (see 3.4.3 and 4.4.1.3) Critical defect - Code No. 03003 - A sample of activator weighing a minimum of 0.1 gram shall be selected from each activator batch prepared to produce the first article sample. The sample shall be tested as specified in 4.4.1.3 using test equipment specified in 4.3.4. The first article sample shall be rejected if it contains an activator batch that is classed defective.

4.3.3.1.1.4 Activator acid resistance (see 3.4.4 and 4.4.1.4) Critical defect - Code No. 03004 - A sample of activator weighing a minimum of 0.1 gram shall be selected from each activator batch prepared to produce the first article sample. The sample shall be tested as specified in 4.4.1.4 using test equipment as specified in 4.3.4. The first article sample shall be rejected if it contains an activator batch that is classed defective.

4.3.3.1.2 Cured Pm-147 activated self-luminous paint.

4.3.3.1.2.1 Wipe test for radiological contamination from cured self-luminous paint (see 3.5.1 and 4.4.2.1) Critical defect - Code No. 04001 - Five samples from each self-luminous paint batch used to produce the first article sample shall be prepared and tested as specified in 4.4.2.1 using test equipment specified in 4.3.4. The first article sample shall be rejected if it contains a paint batch that is classed defective.

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4.3.3.1.2.2 Solubility of cured self-luminous paint (see 3.5.2 and 4.4.2.2) Critical defect - Code No. 04002 - Each of the five samples from the wipe test from each self-luminous paint batch as specified in paragraph 4.3.3.1.2.1 shall be tested as specified in 4.4.2.2 using test equipment specified in 4.3.4. The first article shall be rejected if it contains a paint batch that is classed defective.

4.3.3.1.2.3 Level of Radioactive material in cured self-luminous paint (see 3.5.3 and 4.4.2.3) Critical defect - Code No. 04003 - The test shall be performed 100 percent on all sight components during first article sample production after the satisfactory completion of all other inspections and tests. The sample shall be tested as specified in 4.4.2.3 using test equipment specified in 4.3.4. The laminations of the entire first article sample of completed (or finished) sights shall be separated and the activated self-luminous paint tested for the specified requirement. If one or more sights in the entire first article sample contains radioactive material in excess of the specified requirement, that sample shall be classed defective and the first article sample shall be rejected.

4.3.3.1.3 Complete sight performance and safety.

4.3.3.1.3.1 Phosphorescent color of the complete sight (see 3.7.1) Minor defect - Code No. 05001. - This test shall be performed 100 percent on all sights included in the first article sample. The sample shall be tested as specified in 4.4.3.1 using test equipment specified in 4.3.4. The first article sample shall be rejected if one or more sights are classed defective.

4.3.3.1.3.2 Brightness of the complete sight (see 3.7.2) Major defect - Code No. 05002 - This test shall be performed 100 percent on all sights included in the first article sample. The sample shall be tested as specified in 4.4.3.2 using test equipment specified in 4.3.4. The brightness measurements shall be recorded and identified with the individual sight for testing in accordance with 4.3.3.1.4. The first article sample shall be rejected if one or more sights are classed defective.

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4.3.3.1.3.3 Radiation contamination of the complete sight (see 3.7.3) Critical defect - Code No. 05003 - This test shall be performed 100 percent on all sights included in the first article sample. The sample shall be tested as specified in 4.4.3.3 using test equipment specified in 4.3.4. The first article sample shall be rejected if one or more sights are classed defective.

4.3.3.1.3.4 Complete sight radiation level (see 3.7.4) Critical defect - Code No. 05004 - This test shall be performed 100 percent on all sights included in the first article sample. The test shall be performed as specified in 4.4.3.4 using test equipment specified in 4.3.4. Visual inspection of the sight shall be performed 100 percent for the complete sight radiation test mark. The first article sample shall be rejected if one or more sights are classed defective or if the complete sight radiation test mark is not present on one or more sights.

4.3.3.1.4 Environmental conditioning and survival tests (see 3.8) Water leakage - Critical defect - Code No. 06001; Damage not exposing self-luminous paint - Major defect - Code No. 06002; Damage exposing self-luminous paint - Critical defect - Code No. 06003; Brightness decay in excess of requirement - Major defect-Code No. 06004; Radiation contamination in excess of requirements-Critical defect - Code No. 06005.-This test shall be performed 100 percent on all sights included in the first article sample. The sample shall be subjected to all of the environmental testing of paragraph 4.4.4 followed by visual inspection and testing as specified in 4.4.3.3 and 4.4.3.2 as follows. The paragraph 4.4.3.2 brightness decay measurement shall be performed on the luminous paint on each sight at regular intervals for four months following the initial measurement as performed in 4.3.3.1.3.2. All of the brightness measurements for each of the sights shall be plotted against a time base and the curve extrapolated from the measurements shall be extended to 28 months. The first article sample shall be rejected if one or more sights are classed defective. The sight shall be classed defective if any of the following conditions occur:

a. The sight leaks during the waterproofness test (see 4.4.4.4) Code No. 06001.

b. The sight fails to comply with the damage visual inspection requirement and the self-luminous paint is not exposed (see 3.8.1) Code No. 06002.

c. The sight fails to comply with the damage visual inspection requirement and the self-luminous paint is exposed (see 3.8.1) Code No. 06003.

d. The sight fails to comply with the brightness decay portion of the complete sight brightness requirement (see 3.7.2.1 and 4.4.3.2) Code No. 06004. If the projected brightness of a sight at 28 months does not comply with the requirement the sight shall be classed defective.

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e. The sight fails to comply with the radiation contamination requirement (see 3.7.3 and 4.4.3.3) Code No. 06005.

4.3.3.2 Production

4.3.3.2.1 Activator

4.3.3.2.1.1 Activator specific activity level (see 3.4.1 and 4.4.1.1) Major defect - Code No. 03001 - A 0.3 gram minimum sample of activator shall be selected from each activator batch. The sample shall be tested as specified in 4.4.1.1 using test equipment specified in 4.3.4. The activator batch shall be rejected if the activator batch is classed defective.

4.3.3.2.1.2 Activator solubility (see 3.4.2 and 4.4.1.2) Critical defect - Code No. 03002 - Five samples weighing between 0.02 grams to 0.10 grams shall be selected from each activator batch. The sample shall be tested as specified in 4.4.1.2 using test equipment specified in 4.3.4. The activator batch shall be rejected if the activator batch is classed defective.

4.3.3.2.1.3 Activator temperature stability (see 3.4.3 and 4.4.1.3) Critical defect - Code No. 03003 - A sample weighing a minimum of 0.1 gram shall be selected from each activator batch. The sample shall be tested as specified in 4.4.1.3 using test equipment specified in 4.3.4. The activator batch shall be rejected if the activator batch is classed defective.

4.3.3.2.1.4 Activator acid resistance (see 3.4.4 and 4.4.1.4) Critical defect - Code No. 03004 - A sample of activator weighing a minimum of 0.1 gram shall be selected from each activator batch. The sample shall be tested as specified in 4.4.1.4 using test equipment specified in 4.3.4. The activator batch shall be rejected if the activator batch is classed defective.

4.3.3.2.2 Cured Pm-147 activated self-luminous paint.

4.3.3.2.2.1 Wipe test for radiological contamination from cured self-luminous paint (see 3.5.1 and 4.4.2.1) Critical defect - Code No. 04001 - Five samples shall be prepared from the self-luminous paint from each sight lot. The samples shall be prepared and tested as specified in 4.4.2.1 using test equipment specified in 4.3.4. If the paint is classed defective, the sight lot shall be rejected.

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4.3.3.2.2.2 Solubility of cured self-luminous paint (see 3.5.2 and 4.4.2.2) Critical defect - Code No. 04002 - Each of the five samples from the wipe test of the sight lot as specified in 4.3.3.2.2.1, shall be tested as specified in 4.4.2.2 using test equipment specified in 4.3.4. If the paint is classed defective, the sight lot shall be rejected.

4.3.3.2.2.3 Level of radioactive material in cured self-luminous paint (see 3.5.3 and 4.4.2.3) Critical defect - Code No. 04003 - The sample shall be selected from each sight lot using an AQL of .015 percent and tested as specified in 4.4.2.3 using test equipment specified in 4.3.4. Sight assembly components that contain radioactive material in excess of the requirement shall be classed defective. If one or more defective sights are included in the sample, the sight lot shall be rejected.

4.3.3.2.3 Complete sight performance and safety.

4.3.3.2.3.1 Phosphorescent color of the complete sight (see 3.7.1 and 4.4.3.1) Minor defect - Code No. 05001 - A sample of 10 complete sights shall be selected from each sight lot and tested as specified in 4.4.3.1 using test equipment specified in 4.3.4. The sight lot shall be rejected if one or more sights in the sample is classed defective.

4.3.3.2.3.2 Brightness of the complete sight (see 3.7.2 and 4.4.3.2) Major defect - Code No. 05002 - A sample of 25 complete sights shall be selected from each sight lot and tested as specified in 4.4.3.2. The sight lot shall be rejected if one or more sights in the sample is classed defective.

4.3.3.2.3.3 Radiation contamination of the complete sight (see 3.7.3 and 4.4.3.3) Critical defect - Code No. 05003 - A sample of complete sights shall be selected from each sight lot. The sample shall be selected using an AQL of 0.015 percent. The sample shall be tested as specified in 4.4.3.3 using test equipment specified in 4.3.4. The sight lot shall be rejected if one or more sights is classed defective.

4.3.3.2.3.4 Complete sight radiation level (see 3.7.4 and 4.4.3.4) Critical defect - Code No. 05004 - This test shall be performed on completed sights 100 percent. The test shall be performed as specified in 4.4.3.4 using test equipment specified in 4.3.4. Visual inspection of the sight shall be performed 100 percent for the presence of the complete sight radiation test mark. Defective sights and sights on which a radiation test mark is not present shall be removed from the lot.

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4.3.3.2.4 Environmental conditioning and survival tests (see 3.8 and 4.4.4) Water leakage - Critical defect - Code No. 06001; Damage not exposing self-luminous paint.-Major defect - Code No. 06002; Damage exposing self-luminous paint - Critical defect - Code No. 06003; Radiation contamination in excess of requirement - Critical defect - Code No. 06005.-A sample of 25 complete sights shall be tested as specified in paragraph 4.4.4.4 followed by visual inspection and testing as specified in 4.4.3.3. The sight lot shall be rejected if there are one or more defective sights in the entire sample. The sight shall be classed defective if any of the following conditions occur:

a. The sight leaks during the waterproofness test (see 4.4.4.4) Code No. 06001.

b. The sight fails to comply with the damage visual inspection requirement and the self-luminous paint is not exposed (see 3.8.1) Code No. 06002.

c. The sight fails to comply with the damage visual inspection requirement and the self-luminous paint is exposed (see 3.8.1) Code No. 06003.

d. The sight fails to comply with the radiation contamination requirement (see 3.7.3 and 4.4.3.3) Code No. 06005.

4.3.3.2.5 Brightness decay (see 3.7.2.1) Major defect - Code No. 06004.-A sample of 10 complete sights shall be selected 10 days maximum after production of the lot and tested as follows: The brightness of the self-luminous paint shall be measured at the time of sample selection and at regular one month intervals for the following four months. All of the brightness measurements for each of the sights shall be plotted against a time base and the curve extrapolated from the measurements shall be extended to 28 months. If the projected brightness of a sight at 28 months does not comply with the requirements the sight shall be classed defective. The lot shall be rejected if one or more sights are classed defective.

4.3.4 Inspection Equipment - Equipment Tabulation Number ET-9234281 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Activator

4.4.1.1 Activator specific activity level. - The sample shall be weighed and the radioactivity determined utilizing equipment sufficiently sensitive to detect 0.01 percent of the Pm-147. The specific activity and weight shall be recorded and as established on this record shall serve as the standard for the activator solubility, temperature resistance and acid resistance tests to follow. The activator batch shall be classed defective if it fails to radiate the specific activity specified.

4.4.1.2 Activator solubility - Each of the specified samples shall be soaked at 25 degrees C for 24 hours in 100 milliliters of one of the following solvents: 0.1 normal hydrochloric acid, 5 percent sodium chloride solution, trichloroethylene, toluene and methyl ethyl ketone. Each of the solutions shall be agitated thoroughly not less than five times during the soaking period. A sample of each of the supernatant liquids shall be withdrawn and the radioactivity determined. The radioactivity counting procedure shall be sufficiently sensitive to detect 0.01 percent solubility of the Pm-147. Under the above conditions, activator batches which show a solubility in any of the solvents in excess of the specified level shall be classed defective.

4.4.1.3 Activator temperature stability. - The specified sample shall be heated in air and maintained at the specified temperature for one hour. The sample shall then be soaked at 25 degrees C for 24 hours in 100 milliliters of 0.1 normal hydrochloric acid. The solution shall be agitated thoroughly not less than five times during the soaking period. A sample of the supernatant liquid shall be withdrawn and the radioactivity determined. The radioactivity counting procedure shall be sufficiently sensitive to detect 0.01 percent solubility of the Pm-147. Under the above conditions, activator batches that show a solubility in excess of the specified level shall be classed defective.

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4.4.1.4 Activator acid resistance. - The specified sample shall be soaked for a minimum of 24 hours in 100 milliliters of 6 normal hydrochloric acid at 25 degrees C. The sample shall be agitated thoroughly at least five times during this period. A sample of the supernatant liquid shall then be withdrawn and the radioactivity determined. The radioactivity counting procedure shall be sufficiently sensitive to detect 0.01 percent solubility of the Pm-147. Under the above conditions, activator batches that show a solubility in excess of the specified level shall be classed defective.

4.4.2 Cured Pm-147 activated self-luminous paint.

4.4.2.1 Wipe test for radiological contamination from cured self-luminous paint.-The sample of the Pm-147 paint shall be applied as a one inch diameter disk in the center of a two inch by two inch metal surface. The sample shall be applied using the same method used to apply the paint to the sight and curing shall be the same as that performed on the paint on the sight. Any protective overcoats of varnish or non-radioactive paint which are applied over the self-luminous paint on the sight shall also be applied over the painted disk. The disk shall be wiped ten wipes minimum with a moistened 1/2 inch sponge and the amount of Pm-147 removed determined by beta counting the radioactivity of each of the sponges used. The paint shall be classed defective if the radiation level of the paint removed exceeds the requirement.

4.4.2.2 Solubility of cured self-luminous paint.-The radiation level of each of the samples of Pm-147 paint applied to the metal surface square shall be measured and recorded. Each of the metal surface squares shall be submerged in 100 milliliters of each of the following solutions: water, 5 percent sodium chloride solution, 0.01 normal hydrochloric acid, silicone oil, and denatured ethanol. The samples shall remain completely submersed in these solutions for 24 hours at 25 degrees C. A sample of the supernatant liquid shall be withdrawn and the radioactivity of each sample determined by beta counting or some equivalent method. The paint shall be classed defective if the Pm-147 content of one or more of the supernatant liquids exceeds the requirement.

4.4.2.3 Level of Radioactive material in cured self-luminous paint. - This test shall be performed prior to the application of any protective overcoats of varnish or non-radioactive material which are applied over the self-luminous paint of the sight. This test shall be performed by radiation measurements taken in an area removed from the self-luminous paint application area so that there is no possibility of additional paint contamination to the sight components following the test and prior to sealing of the sight components. The paint shall be classed defective if the level exceeds the specified requirement.

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4.4.3 Complete sight performance and safety.

4.4.3.1 Phosphorescent color of the complete sight.-The self-luminous paint marking color of each finished sight test sample shall be compared visually with source and filter combinations yielding the dominant wavelengths. When filter-source combinations are too bright to permit convenient matching, a neutral filter may be used. Evidence of phosphorescent color values not within the limits specified shall constitute failure of this test. If the complete sight under test fails the test the sight shall be classed defective.

4.4.3.2 Brightness of the complete sight.-The brightness of the self-luminous paint of the finished sight shall be measured using either of the following alternative brightness measurement methods. The measurement shall be recorded identifiable to the sight from which it was measured.

4.4.3.2.1 Comparison method of brightness measurement.-The brightness values of the self-luminous paint of the complete sight shall be determined by a comparison method using photoelectric photometry. The self-luminous paint shall be compared for brightness with the factory standard markings. The factory standard markings shall be the same size and color as the luminous paint markings and the factory standard markings shall have been measured by visual photometry at a Government laboratory or a Government approved testing facility within the last six months prior to performing the test. Prior to performing any measurements, the photoelectric photometer shall be calibrated with a factory standard marking in a dark room. The complete sight shall not be exposed to any light during the specified darkness period or during the test. The meter deflections shall be measured and calculated on each complete sight test sample. If the complete sight fails to comply with the specified requirement the sight shall be classed defective.

4.4.3.2.2 Visual photometer method of brightness measurement.-The self-luminous paint of the complete sight shall be measured for brightness values with a visual photometer. The measurements shall be made by an operator who has been dark-adapted for not less than 30 minutes prior to testing. The complete sight shall not be exposed to any light during the specified darkness period or during the test. If the complete sight fails to comply with the specified requirement the sight shall be classed defective.

4.4.3.3 Radiation contamination of complete sight.-The radiation contamination of the complete sight shall be determined by performing each of the following tests in the stated sequence.

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4.4.3.3.1 Water submersion contamination test.-Each finished sight test sample shall be submerged in the least amount of water possible at room temperature for a period of 24 hours. The water sample shall then be analyzed for its radioactive content by liquid scintillation counting. Disintegrations in excess of the requirement shall constitute failure of the tests. If the complete sight fails the test the sight shall be classed defective.

4.4.3.3.2 Wipe test for contamination.-The entire accessible area of each sight test sample shall be wiped with filter paper. The filter paper shall then be analyzed for radioactive content. If the radioactive content on the paper exceeds the specified maximum, the sight under test shall be classed defective.

4.4.3.4 Radiation level of complete sight.-The radiation level shall be measured over each of the external surfaces of the completed sight. The test shall be performed in an area in which the background radiation is less than 0.4 milliroentgens per hour. Evidence of radiation originating from the completed sight in excess of the requirement shall constitute failure of this test and the sight shall be classed defective. The test equipment shall automatically stamp the complete sight radiation level test mark only on those sights which have been subjected to the test and have not been classed defective.

4.4.4 Environmental conditioning tests of complete sight.- Each complete sight in the entire sample shall be subject to the following tests in the sequence shown without cleaning or wiping of the sights during or following testing.

4.4.4.1 Temperature extremes.-Each complete sight shall be subjected to the temperature conditions in accordance with MIL-E-5272, procedure I.

4.4.4.2 Accelerated weathering.-Each complete sight test sample shall be subjected to 100 hours of accelerated weathering in accordance with Federal Test Method Standard No. 141 Method 6151. The maximum temperature of 150° F shall not be exceeded during the test.

4.4.4.3 Shock - Each complete sight test sample shall be subjected to a shock test in accordance with National Bureau of Standards Handbook No. 66.

4.4.4.3.1 Drop test - Each complete sight test sample shall be dropped upon a concrete or iron surface in a 3 foot gravitational fall, or shall be subjected to equivalent treatment in a test device simulating such a free fall. The drop test shall be repeated 100 times from random orientations.

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4.4.4.3.2 Vibration.-Each complete sight test sample shall be attached to a vibrating fixture and vibrated at a rate of not less than 26 cycles per second and a vibration acceleration of not less than 2 g for a period of not less than one (1) hour.

4.4.4.4 Waterproofness.-Each complete sight test sample shall be immersed in 30 inches of water for not less than 24 hours. The pressure of the air above the water shall then be reduced to 1 inch of mercury absolute pressure. The lower pressure shall be maintained for 1 minute or until air bubbles cease to be given off by the water, whichever is the longer time. The pressure shall then be returned to normal atmospheric pressure and the sight air dried. Evidence of air bubbles coming from within the sight shall cause the sight to be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging - Apply a protective wrapper to entire length of transparent portion of each sight when shipped. Wrapper shall be made from chemically neutral paper conforming to Specification MIL-P-17667, Type I and secured with tape meeting the requirements of Specification UU-T-106, Type I or II.

5.2 Packing - Sights wrapped as specified in 5.1 shall be packed in accordance with MIL-STD-1169. A maximum of forty (40) units of radiation shall be permitted for shipment in one vehicle or storage in one location in accordance with Code of Federal Regulations '49, '71-90'. A unit of radiation is defined by the Federal Regulation as being equal to one milliroentgen per hour of gamma radiation or equivalent at a distance of one meter from any accessible external surface of the package.

5.3 Marking - Marking shall be in accordance with MIL-STD-1169 and each shipping container shall have the following precautionary label:

Radioactive Material label, showing milliroentgens per hour of radiation at surface of container and milliroentgens per hour of radiation when measured at a distance of one (1) meter from container.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. All submitted designs shall contain a reference to the applicable 5 digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to

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fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND14. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article samples. - Instructions as to location for the evaluation of the first article sample, shall be obtained from the contracting officer. Upon receipt of such request the contracting officer shall advise Picatinny Arsenal, and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, N.J. ATTN: SMUPA-NI.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, fur-

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Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1340

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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COMMANDING OFFICER
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Picatinny Arsenal
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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